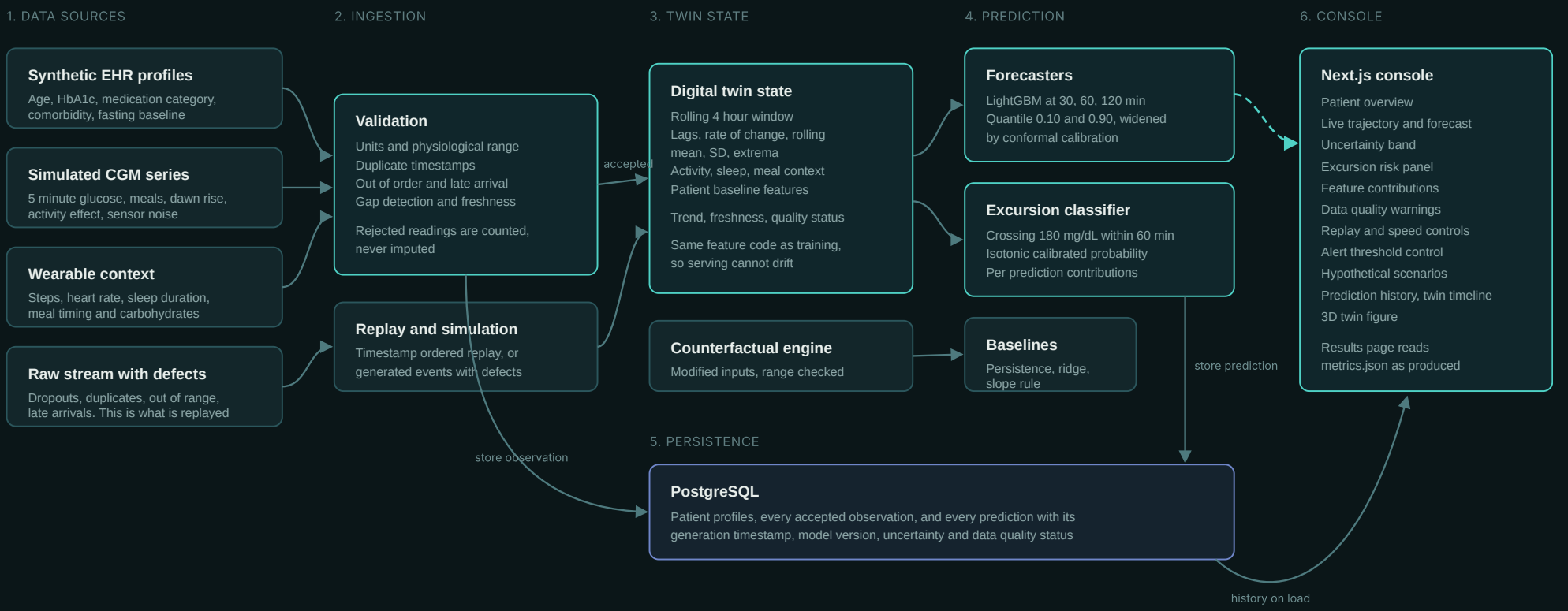
GlycoTwin system architecture

Patient digital twin for glucose forecasting and excursion detection, Type 2 diabetes

Research prototype. Synthetic data only. Not a medical device.



Request or write

WebSocket push to the console

Component carrying model or validation logic

Persistence

Pipeline for every sensor event

1. Receive the event with units and source metadata
2. Validate identifier, timestamp, units and physiological range
3. Handle duplicate, out of order, late and missing readings
4. Update the rolling feature window and twin state
5. Run the forecasters and the excursion classifier
6. Persist the prediction with its generation timestamp
7. Publish the updated result to the console

Six components as specified in competition.txt section 11. Stack: FastAPI, LightGBM, PostgreSQL, WebSockets, Next.js, React Three Fiber, Docker Compose.

glycotwin-0.1.0